An integrated biasing power supply, control and acquisition system for large Langmuir probe arrays in cold plasmas

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***Abstract*—Large Langmuir probe arrays are widely used in plasma-facing diagnostics for fusion devices such as MITICA, the full-scale neutral beam injection prototype for ITER, where robust and synchronized measurement systems are required. This work presents an integrated power supply, control and data acquisition system for large Langmuir probe arrays operating in cold plasmas, designed for compatibility with the ITER Control, Data Access and Communication framework. The proposed system is based on a modular architecture using automotive-grade components and System-on-Modules that integrate Field Programmable Gate Arrays fabric and ARM processing cores, enabling flexible real-time control and data processing. A custom Linux distribution based on PetaLinux, combined with the MARTe2 framework, supports real-time monitoring and processing of probe signals. The system is implemented in a single rack and supports floating-potential, single-probe and double-probe measurement configurations. The analog front-end is designed to operate over a wide range of plasma conditions relevant to MITICA, with electron temperatures in the range of 1–10 eV. A distributed biasing unit integrated within each probe module performs the full polarization cycle locally, eliminating the need for external bias supplies and improving system scalability. Time synchronization across distributed diagnostics is achieved using either a dedicated timing system or the Precision Time Protocol (PTP).**

***Index Terms*—Acquisition Systems, Analogue Electronics, Digital Electronics, FPGAs, Fusion, Langmuir Probes**

I. Introduction

The characterization of plasma parameters is essential for understanding and optimizing plasma-based systems. Among the available diagnostic techniques, Langmuir probes are widely used because they provide direct access to key plasma properties, including electron density, electron temperature, plasma potential, and the electron energy distribution function [1], [2]. These quantities are extracted from the probe current–voltage (I–V) characteristic, making Langmuir probes a versatile and effective diagnostic tool across a broad range of plasma conditions. However, due to their intrusive nature, their application is generally limited to low-temperature plasmas, where the perturbation introduced by the probe remains acceptable.

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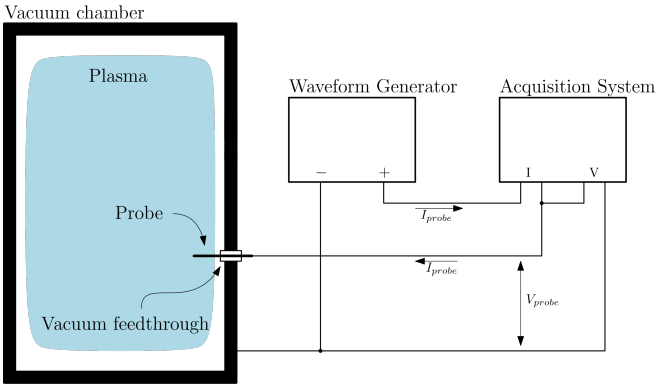


Figure 1: General layout of a Langmuir probe acquisition system.

Fig. 1 illustrates the general configuration of a Langmuir probe measurement system. The diagnostic principle is conceptually simple: a metallic electrode is inserted into the plasma, biased over a controlled voltage range, and the resulting current response is measured. In practice, the setup typically includes a programmable bias power supply and a data acquisition system for recording the I–V characteristic. Such systems are commonly implemented using commercial off-the-shelf (COTS) instrumentation and are well suited for applications involving a limited number of probes. However, when a large number of spatially distributed measurements is required, conventional architectures become increasingly difficult to scale. The resulting increase in system complexity, footprint, and cost limits their applicability in large-area plasma diagnostics.

A representative example of large-scale cold-plasma diagnostics is the Megavolt ITER Injector and Concept Advancement (MITICA), shown in Fig. 2. MITICA is the full-scale prototype of the neutral beam injector (NBI) for the International Thermonuclear Experimental Reactor (ITER) [3]. Installed at the PRIMA test facility in Padua, Italy, it is designed to demonstrate full-performance operation of the ITER neutral beam system, including the generation and acceleration of negative ion beams up to 1 MeV. Due to the large physical dimensions and complexity of the injector, plasma monitoring requires multiple diagnostics distributed across several subsystems.

As shown in Fig. 2, Langmuir probes are installed in several

Figure 2: MITICA design highlighting Langmuir probe locations [3].

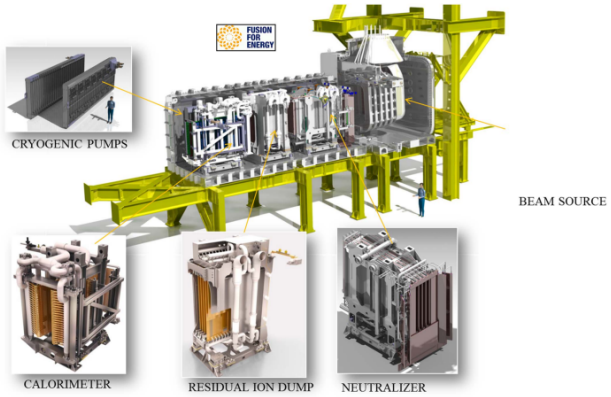


Table I: Measurement ranges and relative resolution requirements.

Measurement Range	Minimum nominal resolution	Maximum residual r.m.s. noise
Voltage measurements		
$\pm 120\,\mathrm{V}$	$50\,\mathrm{mV}$	$15\,\mathrm{mV}$
Current measurements		
$\pm 1\,\mathrm{mA}$	$0.5\,\mu\mathrm{A}$	$0.15\,\mu\mathrm{A}$
$\pm 10\,\mathrm{mA}$	$5\,\mu\mathrm{A}$	$1.5\,\mu\mathrm{A}$
$\pm 100\,\mathrm{mA}$	$50\,\mu\mathrm{A}$	$1.5\,\mu\mathrm{A}$
$\pm 1\,\mathrm{A}$	$0.5\,\mathrm{mA}$	$0.15\,\mathrm{mA}$

regions of the acceleration chain, resulting in a total of 212 measurement points: 62 in the Neutralizer, 48 in the Residual Ion Dump (RID), and 102 in the Calorimeter. Managing such a large number of spatially distributed probes using conventional acquisition architectures is impractical, motivating the development of a dedicated and scalable solution.

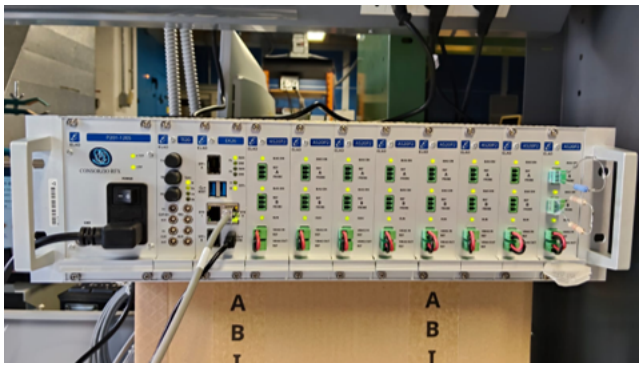
To address these challenges, this work presents EPICA (Electrostatic Probe Integrated Conditioning and Data Acquisition module), an integrated platform that combines power supply, control, and data acquisition functionalities for large Langmuir probe arrays. Unlike conventional systems based on centralized COTS instrumentation, EPICA embeds the probe biasing stage directly within each acquisition module. This approach improves scalability, reduces wiring complexity, and increases modularity, making the system better suited for high-density probe deployments.

EPICA is built on a high-performance hardware and software framework fully compatible with the ITER Control, Data Access and Communication (CODAC) system. The platform is designed to operate across a wide range of plasma conditions, supporting electron densities from fractions of $10^{15}\,\mathrm{m}^{-3}$ to $10^{18}\,\mathrm{m}^{-3}$, multiple probe configurations, and real-time data processing.

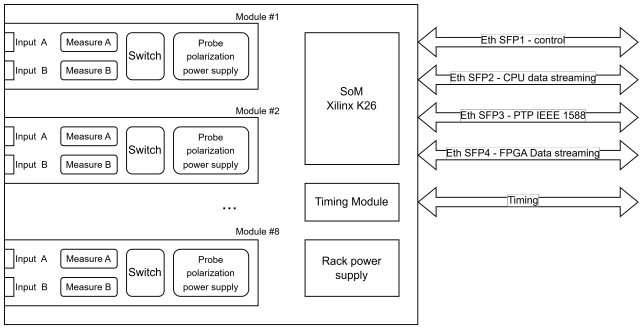
The proposed architecture supports synchronized operation across multiple distributed racks while integrating probe bias

Figure 3: EPICA rack and system overview.

(a) EPICA rack: from left to right the rack power supply, the timing module, the control module and eight probe modules.



(b) Epica internal system overview.



generation, signal conditioning, and data acquisition within each module. This significantly improves scalability and reduces system-level complexity while satisfying the measurement resolution and noise requirements summarized in Table I.

The main contributions of this work are threefold: (i) the development of a fully integrated probe-level biasing and acquisition module for high-density Langmuir probe arrays; (ii) a modular rack-based architecture scalable to large numbers of probes while maintaining synchronization and centralized control; and (iii) a hardware and software implementation based on Field Programmable Gate Arrays (FPGA), System-on-Module (SoM) technology, and real-time frameworks enabling deterministic acquisition and processing in fusion-relevant plasma environments.

The remainder of this paper is organized as follows. Section II describes the EPICA system architecture; Section III details the integrated polarization and acquisition module; Section IV presents the software implementation; Section V reports the experimental validation; Section VI discusses the results; and Section VII concludes the paper.

II. EPICA ARCHITECTURE

EPICA adopts a modular architecture designed to ensure scalability and portability. Fig. 3a shows the rack enclosure hosting the complete system, whereas Fig. 3b illustrates the internal system overview. The platform supports up to 16 probes, arranged into eight electrically and functionally independent probe pairs. Each pair is managed by a dedicated acquisition

and biasing module responsible for electrode polarization and signal acquisition. This modular organization enables straightforward system expansion while maintaining electrical isolation and operational flexibility, making EPICA suitable for a broad range of plasma diagnostics applications.

The system consists of four main subsystems: the power-supply unit, timing module, controller module and probe modules.

The power-supply unit generates the primary $5\,\mathrm{V}$ and $12\,\mathrm{V}$ DC rails distributed across the modules. Additional voltage levels required by individual subsystems are generated locally through dedicated DC/DC converters integrated within each module. This distributed power architecture improves electrical isolation and mitigates ground-loop effects.

The timing module provides the synchronization reference for both the controller and probe modules. It can operate either as a standalone timing source or be locked to an external master clock, enabling synchronization with other diagnostics and consistent temporal correlation of measurements. Alternatively, when the Precision Time Protocol (PTP) is adopted [4], synchronization is distributed through the Ethernet network, allowing the timing module to be bypassed.

To support integrated probe biasing, real-time data streaming and high-resolution acquisition, EPICA relies on a high-performance controller platform based on the Xilinx K26 System-on-Module (SoM) [5]. The selected platform combines a Field-Programmable Gate Array (FPGA) with an ARM embedded processing system, enabling tight integration between deterministic real-time control and high-throughput data processing.

The system software is implemented using the MARTe2 real-time framework, executed on the embedded processor to manage supervisory control, data acquisition and communication tasks. Time-critical functions, including low-latency signal acquisition and preprocessing, are implemented within the FPGA. This hardware-software partitioning ensures deterministic real-time performance while preserving flexibility for high-level control and data management.

The probe modules implement the analog front-end electronics, probe bias generation, signal conditioning and analog-to-digital conversion. Digitized measurements are transmitted to the controller for further processing, storage and streaming.

Communication between EPICA and external systems is performed through Ethernet interfaces available on the controller as illustrated in Fig. 3b, enabling simultaneous data streaming, synchronization and integration with supervisory frameworks such as ITER CODAC. In addition, a USB serial console is provided for system configuration, firmware deployment and debugging.

III. Integrated Polarization and Acquisition Module

This section describes the architecture of the probe module shown in Fig. 4, highlighting the design choices adopted to achieve compactness, flexibility and robustness in high-density Langmuir probe systems.

By embedding functionalities traditionally implemented through distributed laboratory equipment, as illustrated in Fig. 1, the system significantly reduces spatial overhead and improves scalability for large arrays.

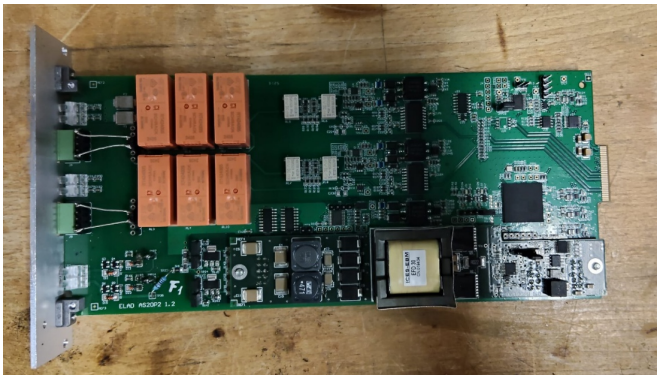


Figure 4: Integrated probe management module, including biasing and signal conditioning stages.

A key feature of the architecture is its ability to dynamically reconfigure both measurement range and probe measurement topology without requiring physical intervention at the instrumentation rack. To ensure robustness under harsh operating conditions, the plasma-facing circuitry is galvanically isolated from the digital processing electronics. This isolation strategy protects low-voltage hardware while preserving signal integrity in the presence of high-voltage transients and electromagnetic interference (EMI).

The module supports a programmable polarization voltage up to $\pm 120\,\mathrm{V}$, with a maximum drive current of $800\,\mathrm{mA}$. Signal acquisition is performed through a 16-bit analog-to-digital conversion chain with programmable sampling frequency, enabling adaptation to different plasma operating conditions.

A. Analog Front End

The analog front-end provides the electrical interface between the Langmuir probes and the acquisition electronics. Each module supports two independent probe channels, both equipped with dedicated voltage and current sensing circuits to enable simultaneous acquisition of the probe I-V characteristic.

The system supports floating-potential, single-probe and double-probe measurement configuration. An additional configuration is supported to disconnect the probes from the circuit.

To maximize robustness in electrically noisy environments, the front-end is implemented exclusively with passive components at the plasma-facing interface, avoiding active semiconductor devices. This design choice reduces susceptibility to EMI and improves reliability under harsh electrical conditions.

Measurement range selection and probe configuration are implemented through two switch networks that dynamically select shunt resistors and operating modes without introducing active gain stages. Probe potential measurements are performed using a fixed voltage divider dimensioned to withstand the maximum bias voltage range. In addition, dedicated protection circuits are implemented at the input of each acquisition channel to mitigate voltage and current transients.

B. Biasing Power Supply

Conventional COTS power supplies are generally designed for broad laboratory applications, where linear behavior is often a key requirement, and are therefore oversized for Langmuir probe diagnostics, leading to unnecessary cost, complexity and footprint overhead. In contrast, the electrical requirements of Langmuir probes are dictated by the probe I-V characteristic.

In the ion saturation region, where the probe voltage V_p is much smaller than the floating potential V_f, relatively high bias voltages are required while probe current remains limited. Conversely, when $V_p > V_f$, that is, when the probe is in the electron retardation region, current increases approximately exponentially whereas the required voltage is comparatively low. Since maximum voltage and maximum current are not required simultaneously, the power delivery stage can be optimized for the effective operating envelope of the diagnostic.

The bias supply adopts a dual-stage output control architecture consisting of a preregulator followed by a bipolar DC-DC converter. The preregulator provides slow but high-precision voltage control, while the DC-DC stage ensures fast transient response during bias modulation. Output polarity selection is implemented through MOSFET-based switching, allowing the appropriate converter output to be routed to the probe.

Bias waveform generation can be adapted to different measurement scenarios. Although a high-speed communication interface enables real-time waveform updates, closed-loop execution latency may limit performance during high-frequency voltage sweeps. To address this issue, waveform samples can be preloaded into controller memory, reducing communication overhead and enabling deterministic waveform generation.

C. Control and Acquisition Unit

The probe module integrates both an FPGA and a microcontroller, each electrically isolated from the high-voltage bias stage and analog front-end. The FPGA is responsible for time-critical operations, including fast control loops, synchronization and protection logic, while the microcontroller manages supervisory functions such as configuration and monitoring.

Isolation between the analog and digital domains is achieved through sigma-delta ($\Sigma\Delta$) analog-to-digital (ADC) converters with a nominal resolution of 16 bits. The 1-bit modulator operates at a sampling frequency of $f_s = 20$ MHz, while the effective output sampling frequency F_s and effective number of bits (ENOB) depend on the selected oversampling ratio (OSR), according to

$$F_s = \frac{f_s}{\mathrm{OSR}}, \qquad \mathrm{ENOB} = \frac{\mathrm{SNR} - 1.76}{6.02}, \tag{1}$$

where SNR denotes the signal-to-noise ratio. The relationship highlights the inherent trade-off between output sampling rate and effective resolution: increasing the oversampling ratio improves SNR performance (and thus ENOB), at the cost of reduced effective output bandwidth.

IV. Real Time Operative System - RTOS

V. Experimental Setup and System Validation

VI. Experimental Results and Discussion

VII. Conclusion

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